

Pan retinal photocoagulation for proliferative diabetic retinopathy: pattern scan laser versus argon laser.



PURPOSE OF REVIEW: Diabetic retinopathy is the leading cause of visual impairment in working-age adults worldwide. Pan retinal photocoagulation (PRP) has provided an effective treatment to decrease the risk of severe vision loss in patients with proliferative diabetic retinopathy for the past four decades. Pattern scan laser (PASCAL) was developed to minimize the side effects of PRP. The purpose of this review is to discuss the differences between the traditional argon laser and the PASCAL. RECENT FINDINGS: PASCAL can achieve comparable results with the conventional argon PRP in the treatment of patients with diabetic retinopathy. The PASCAL delivery system creates well aligned arrays of retinal lesions in a shorter period. PASCAL provides amore comfortable profile when compared to the argon laser. SUMMARY: The PASCAL is now being substituted for the conventional argon laser for PRP in many clinics. Ophthalmologists should keep in mind that adjusting the PASCAL settings (including the duration, number, and size of laser burns) might become necessary to maintain regression and eliminate recurrence of neovascularization in patients with proliferative diabetic retinopathy. Further studies are needed to determine the parameters for optimal safety and efficacy on the PASCAL.